



Algebra 1

Sample Test Materials

The purpose of these sample test materials is to orient teachers and students to the types of paper-based B.E.S.T. Algebra 1 questions. By using these materials, students will become familiar with the types of items and response formats they may see on a paper-based test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test. The sample test materials are not intended to guide classroom instruction.

B.E.S.T. Algebra 1 EOC Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches
1 yard = 3 feet
1 mile = 5,280 feet
1 mile = 1,760 yards

1 cup = 8 fluid ounces
1 pint = 2 cups
1 quart = 2 pints
1 gallon = 4 quarts

1 pound = 16 ounces
1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters
1 meter = 1000 millimeters
1 kilometer = 1000 meters
1 liter = 1000 milliliters

1 gram = 1000 milligrams
1 kilogram = 1000 grams

Time Conversions

1 minute = 60 seconds
1 hour = 60 minutes
1 day = 24 hours
1 year = 365 days
1 year = 52 weeks

Formulas

Forms of Linear Equations	Forms of Quadratic Functions	Forms of Exponential Functions
$y = mx + b$ $Ax + By = C$ $y - y_1 = m(x - x_1)$	$f(x) = ax^2 + bx + c$ $f(x) = a(x - h)^2 + k$ $f(x) = a(x - p)(x - q)$	$f(x) = ab^x$ $f(x) = a(1 \pm r)^x$

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

where $ax^2 + bx + c = 0$ and $a \neq 0$

Final Amounts under Simple Interest	Final Amounts under Compound Interest
$A = P(1 + rt)$ where P = principal, r = rate, and t = time	$A = P\left(1 + \frac{r}{n}\right)^{nt}$ where P = principal, r = rate, n = number of times compounded, and t = time

Use the space in this Test and Response Book to do your work. Then, completely fill in the bubble beside the answer you choose. For some items, filling in more than one bubble may be required, so read each item carefully. If you change your answer, be sure to erase completely.

Some items will ask you to write a response in a shaded box or boxes. See the sample item below.

Sample Item:

An expression is shown.

$$(22.4)(2.65)$$

What is the value of the expression?

Write your response in the shaded box below.

	Write your response in the box.
--	--

Some items may have more than one box, so read each item carefully. Your answers for the items with response boxes may contain whole numbers, fractions, decimals, or negative numbers.

1. A band sells x premium tickets and y regular tickets for a concert.

- A premium ticket costs \$20.
- A regular ticket costs \$5 less than a premium ticket.
- The band raises \$2145 from selling tickets.

Select coefficients to complete the equation representing the relationship between x and y . For each box, fill in the bubble before the number that is correct.

<table border="1"><tr><td>☐ A</td><td>5</td></tr><tr><td>☐ B</td><td>15</td></tr><tr><td>☐ C</td><td>20</td></tr><tr><td>☐ D</td><td>25</td></tr></table>	☐ A	5	☐ B	15	☐ C	20	☐ D	25	$x +$	<table border="1"><tr><td>☐ A</td><td>5</td></tr><tr><td>☐ B</td><td>15</td></tr><tr><td>☐ C</td><td>20</td></tr><tr><td>☐ D</td><td>25</td></tr></table>	☐ A	5	☐ B	15	☐ C	20	☐ D	25	$y = 2145$
☐ A	5																		
☐ B	15																		
☐ C	20																		
☐ D	25																		
☐ A	5																		
☐ B	15																		
☐ C	20																		
☐ D	25																		

- 2.** A function is shown.

$$f(x) = 2x + 1$$

Select all the effects on the graph of the function when $f(x)$ is multiplied by 3.

- | | | |
|--|--|---|
| ☐ Ⓐ The x -intercept increases. | ☐ Ⓒ The y -intercept increases. | ☐ Ⓔ The slope increases. |
| ☐ Ⓑ The x -intercept decreases. | ☐ Ⓓ The y -intercept decreases. | ☐ Ⓕ The slope decreases. |

3. The expression $15,000(1 + 0.02)^m$ can be used to model the sales, in dollars, of a company after m months.

The value 12 was substituted for m to create the expression $15,000(1 + 0.02)^{12}$.

Write your response in the shaded box below and fill in the bubble before the correct unit to complete the sentence.

The expression $(1 + 0.02)^{12}$ represents the growth factor of the company's sales for a period of time

lasting

- ☐ (A) day(s)
☐ (B) month(s)
☐ (C) year(s)

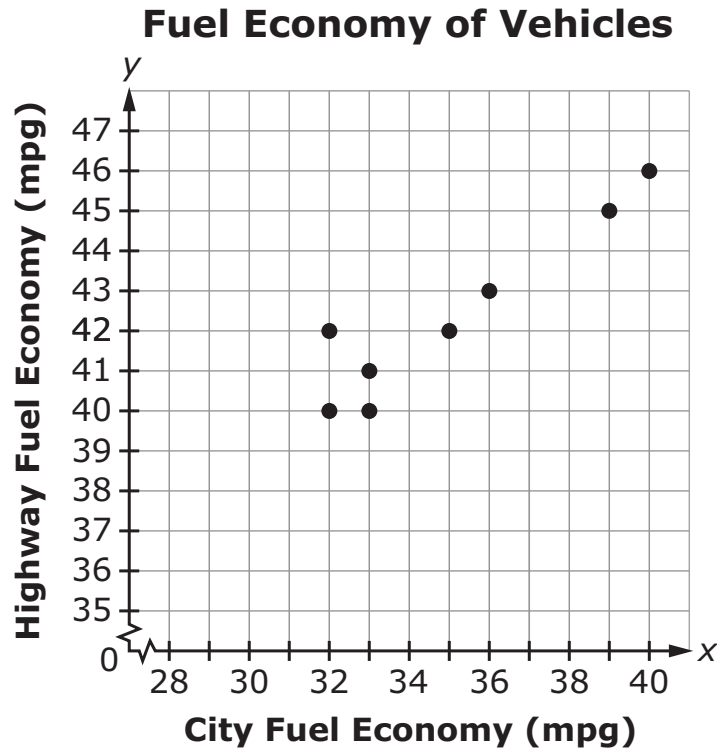
4. A table of values for a linear function is shown.

x	$f(x)$
-1	-8
3	0
6	6

Fill in bubbles to match each key feature of the graph to its value.

	-6	$\frac{1}{2}$	2	3	6
x-intercept	(A)	(B)	(C)	(D)	(E)
y-intercept	(F)	(G)	(H)	(I)	(J)
rate of change	(K)	(L)	(M)	(N)	(O)

5. The fuel economy of a car is its average distance traveled per amount of fuel consumed. Cars have different fuel economies when they are driven in the city and on the highway. A linear model of the relationship between the city and highway fuel economies of eight cars, in miles per gallon (mpg), is shown.



This question has **two** parts.

Part A

Which linear equation best models the scatter plot?

Ⓐ $y = \frac{2}{3}x + 19$

Ⓑ $y = \frac{6}{8}x + 25$

Ⓒ $y = \frac{3}{2}x + 19$

Ⓓ $y = \frac{8}{6}x + 25$

Part B

Based on the model in Part A, what could be the highway fuel economy, in mpg, of a car that has a city fuel economy of 43 mpg? Round your answer to the nearest hundredth.

Write your response in the shaded box below.

6. The table shown represents two functions, Function A and Function B.

x	Function A	Function B
0	700	600
1	750	624
10	1200	888*
20	1700	1315*

*rounded value

Complete the sentences to compare the two functions. For each blank, fill in the bubble **before** the word or phrase that is correct.

Function A appears to grow _____ [☐ linearly ☐ exponentially].

Function B appears to grow _____ [☐ linearly ☐ exponentially].

As x increases, the quantities for _____ [☐ Function A ☐ Function B]

will eventually exceed the quantities for _____ [☐ Function A

☐ Function B].

B.E.S.T. Mathematics Sample Items

7. In a survey of 3,260 people, 57% of people said they spend more than 2 hours a day on their smartphones. The margin of error is $\pm 2.2\%$. The survey is used to estimate the number of people in a town of 17,247 who spend more than 2 hours a day on their smartphones.

Based on the survey, what are the estimated minimum and maximum numbers of people in the town who spend more than 2 hours a day on their smartphones? Round your answers to the nearest whole numbers.

Write your responses in the shaded boxes below.

Minimum:

Maximum:



Algebra 1

Sample Items

Answer Key

The B.E.S.T. Algebra 1 Sample Items Answer Key provides the correct response(s) for each item on the sample test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test.

B.E.S.T. Mathematics Sample Items Answer Key

- 1.** Doug has 8 square pieces of wood. Each piece of wood has a side length of s centimeters. The total area of all 8 pieces of wood is 200 square centimeters.

Create a quadratic equation Doug can use to find the side length of each piece of wood.

$8s^2=200$

← → ↶ ↷ ✖

1	2	3	s								
4	5	6	+	-	•	÷					
7	8	9	<	≤	=	≥	>				
	0		$\square^\square$	$\square_\square$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	i	
.	-	$\frac{\square}{\square}$	sin	cos	tan	arcsin	arccos	arctan			

Other correct responses: any equivalent equation

B.E.S.T. Mathematics Sample Items Answer Key

2. A band sells x premium tickets and y regular tickets for a concert.

- A premium ticket costs \$20.
- A regular ticket costs \$5 less than a premium ticket.
- The band raises \$2145 from selling tickets.

Select coefficients to complete the equation representing the relationship between x and y .

∇ $x +$ ∇ $y = 2145$

B.E.S.T. Mathematics Sample Items Answer Key

3. Enter an expression to make a true equation.

$$(xyz)^2 - 1 = (\text{ } xyz + 1 \text{ }) \cdot (xyz - 1)$$

← → ↶ ↷ ✖													
1	2	3	x	y	z								
4	5	6	+	-	•	÷							
7	8	9	<	≤	=	≥	>						
	0		$\square^\square$	$\square_\square$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	i			
.	-	$\frac{\square}{\square}$	sin	cos	tan	arcsin	arccos	arctan					

Other correct responses: any equivalent expression

B.E.S.T. Mathematics Sample Items Answer Key

4. A function is shown.

$$f(x) = 2x + 1$$

Select all the effects on the graph of the function when $f(x)$ is multiplied by 3.

☐ The x-intercept increases.

☒ The y-intercept increases.

☒ The slope increases.

☐ The x-intercept decreases.

☐ The y-intercept decreases.

☐ The slope decreases.

Option C: **This answer is correct.** The student correctly indicated an effect on the graph with the specified value of k .

Option E: **This answer is correct.** The student correctly indicated an effect on the graph with the specified value of k .

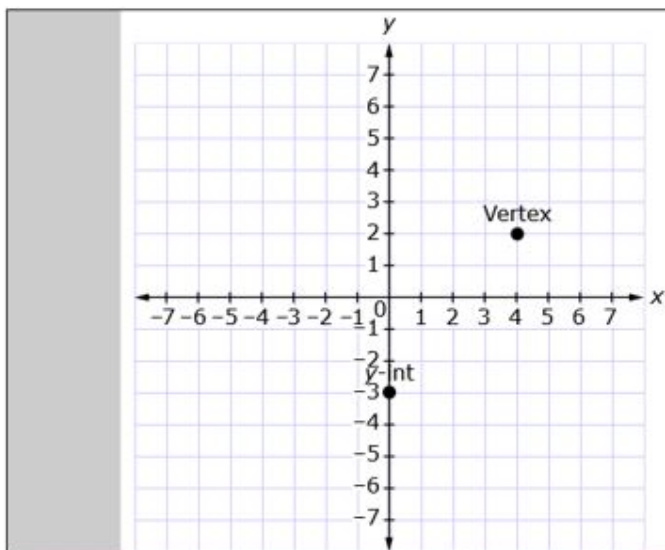
5. The equation for the function f is shown.

$$f(x) = -\frac{1}{2}x + 3$$

The quadratic function g has the following properties:

- g has a lesser y -intercept than f .
- The maximum value and y -intercept of g are not equal.
- First g increases, and then it decreases.

Drag the vertex and y -intercept labels to show possible locations for the vertex and y -intercept of g .



B.E.S.T. Mathematics Sample Items Answer Key

6. The expression $15,000(1 + 0.02)^m$ can be used to model the sales, in dollars, of a company after m months.

The value 12 was substituted for m to create the expression $15,000(1 + 0.02)^{12}$.

Enter a number and select a unit to complete the sentence.

The expression $(1 + 0.02)^{12}$ represents the growth factor of the company's sales for a period of time lasting

Other correct responses:

The expression $(1 + 0.02)^{12}$ represents the growth factor of the company's sales for a period of time lasting

or

The expression $(1 + 0.02)^{12}$ represents the growth factor of the company's sales for a period of time lasting

or

The expression $(1 + 0.02)^{12}$ represents the growth factor of the company's sales for a period of time lasting

B.E.S.T. Mathematics Sample Items Answer Key

7. A table of values for a linear function is shown.

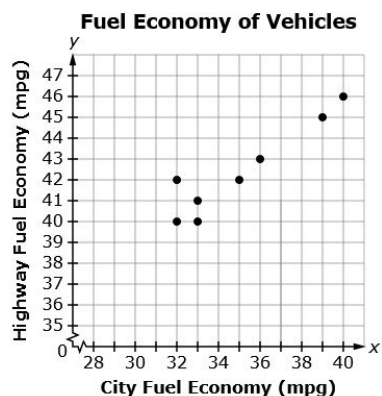
x	$f(x)$
-1	-8
3	0
6	6

Match each key feature of the graph to its value.

	-6	$\frac{1}{2}$	2	3	6
x -intercept	☐	☐	☐	☒	☐
y -intercept	☒	☐	☐	☐	☐
rate of change	☐	☐	☒	☐	☐

B.E.S.T. Mathematics Sample Items Answer Key

8. The fuel economy of a car is its average distance traveled per amount of fuel consumed. Cars have different fuel economies when they are driven in the city and on the highway. A linear model of the relationship between the city and highway fuel economies of eight cars, in miles per gallon (mpg), is shown.



This question has **two** parts.

Part A

Which linear equation best models the scatter plot?

☒ A $y = \frac{2}{3}x + 19$

☐ B $y = \frac{6}{8}x + 25$

☐ C $y = \frac{3}{2}x + 19$

☐ D $y = \frac{8}{6}x + 25$

Part B

Based on the model in Part A, what could be the highway fuel economy, in mpg, of a car that has a city fuel economy of 43 mpg? Round your answer to the nearest hundredth.

47.66



Option A: This answer is correct. The student determined the equation of the linear model.

Other correct responses: for Part B, any value between 47.66 and 47.67, inclusive

B.E.S.T. Mathematics Sample Items Answer Key

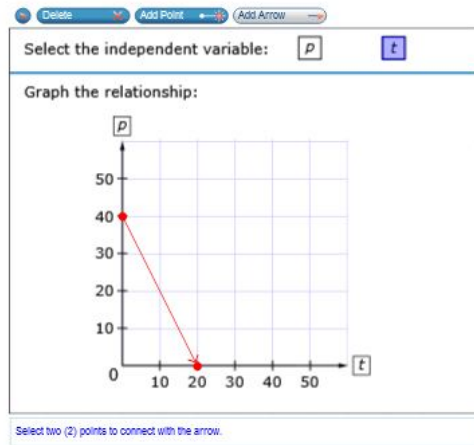
9. This question has **two** parts.

Giselle has 40 prizes for raffle ticket winners. She gives away 2 prizes each day until she runs out of prizes.

The equation $p + 2t = 40$ represents the situation, where p is the number of prizes Giselle has left after t days.

Part A

- First, select p or t to represent the independent variable.
- Then, use the Add Arrow tool to graph the relationship.



Part B

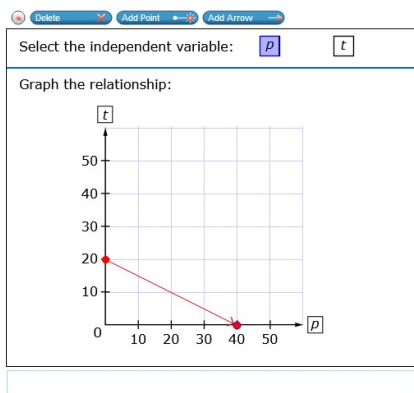
Enter a value in each box to describe the relationship.

The domain of the relationship is from 0 to 20, and Giselle will give away all the prizes after 20 days.

Other correct responses:

Part A

- First, select p or t to represent the independent variable.
- Then, use the Add Arrow tool to graph the relationship.



Part B

Enter a value in each box to describe the relationship.

The domain of the relationship is from 0 to 40, and Giselle will give away all the prizes after 20 days.

B.E.S.T. Mathematics Sample Items Answer Key

- 10.** A school office surveys sixth and seventh grade students about whether they ride the bus to school. The data are recorded in a frequency table.

Enter numbers to complete the frequency table for the data.

	Ride Bus	Do Not Ride Bus	Total
Sixth Grade	176	74	250 
Seventh Grade	96 	54 	150 
Total	272 	128	400

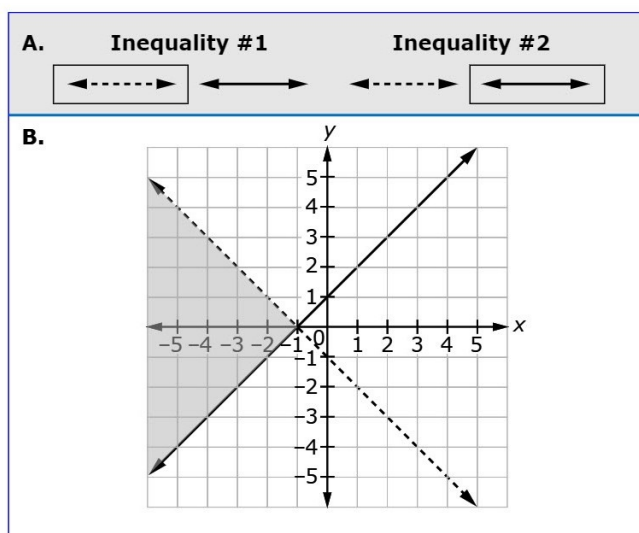
Other correct responses: any equivalent values

- 11.** A system of inequalities is shown.

Inequality #1: $y < -x - 1$

Inequality #2: $y \geq x + 1$

- A. Select a solid or dotted boundary line for each inequality.
- B. Click in a region to graph the solution set of the system.



B.E.S.T. Mathematics Sample Items Answer Key

- 12.** The table shown represents two functions, Function A and Function B.

x	Function A	Function B
0	700	600
1	750	624
10	1200	888*
20	1700	1315*

*rounded value

Complete the sentences to compare the two functions.

Function A appears to grow Function B appears to grow As
x increases, the quantities for will eventually exceed the quantities for

B.E.S.T. Mathematics Sample Items Answer Key

- 13.** In a survey of 3,260 people, 57% of people said they spend more than 2 hours a day on their smartphones. The margin of error is $\pm 2.2\%$. The survey is used to estimate the number of people in a town of 17,247 who spend more than 2 hours a day on their smartphones.

Based on the survey, what are the estimated minimum and maximum numbers of people in the town who spend more than 2 hours a day on their smartphones? Round your answers to the nearest whole numbers.

Minimum:

Maximum:

← → ↶ ↷ ✖		
1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Other correct responses: for the minimum, any value between 9451 and 9452, inclusive; for the maximum, any value between 10210 and 10211, inclusive

B.E.S.T. Mathematics Sample Items Answer Key

- 14.** An expression is given, where $x \neq 0$.

$$\frac{2x^7 + 8x^4 - 14x}{2x}$$

Richard divides incorrectly to get the given quotient.

Click on the incorrect parts of Richard's quotient.

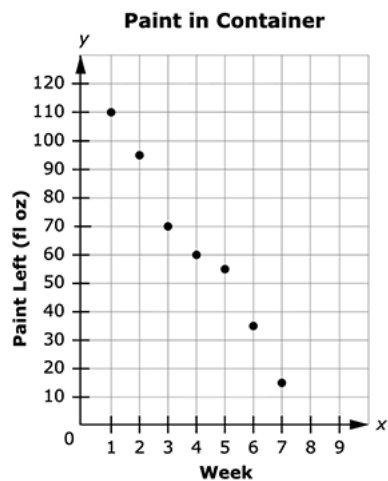
$$2x^6 + 4x^3 - 7x^1$$

B.E.S.T. Mathematics Sample Items Answer Key

- 15.** The number of fluid ounces (fl oz) of paint, y , an artist has left in her paint container after x weeks is shown in the table and the scatter plot.

Paint in Container

Week	1	2	3	4	5	6	7
Paint Left (fl oz)	110	95	70	60	55	35	15



B.E.S.T. Mathematics Sample Items Answer Key

This question has **two** parts.

Part A

Which linear equation **best** models the artist's data?

☐ A $y = -15x + 123$

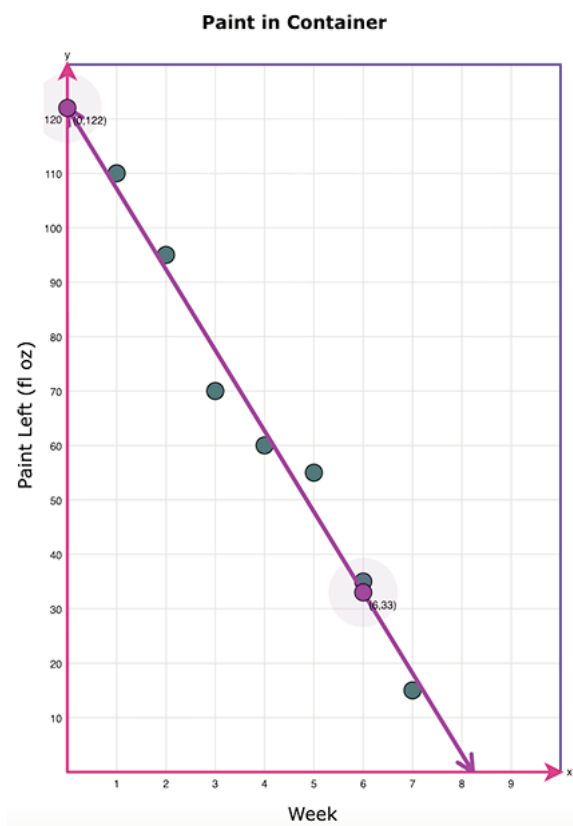
☐ B $x = -15y + 123$

☐ C $y = -95x + 110$

☐ D $x = -95y + 110$

Part B

Plot two points to create a line of fit for the artist's data.



Option A: This answer is correct. The student identified the linear equation that best fits the artist's data in the scatter plot.

Other correct responses: for Part B, any line with a slope between -14.5 and -15.5 and a y -intercept between 121 and 125 , inclusive

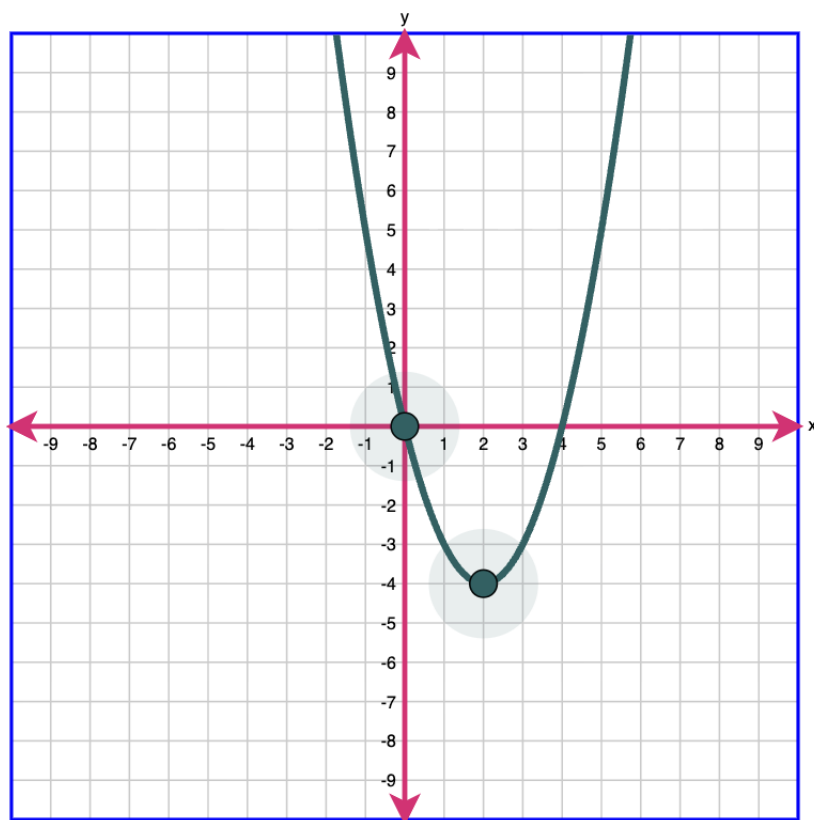
16. A quadratic function is given.

$$f(x) = (x - 2)^2 - 4$$

This question has **two** parts.

Part A

Plot two points to graph the function on the grid.



B.E.S.T. Mathematics Sample Items Answer Key

Part B

What are the x -intercepts of the given function?

$x =$

$x =$

A digital calculator interface with a grid of buttons. The buttons are arranged in a 5x3 grid. The first row contains buttons for 1, 2, and 3. The second row contains buttons for 4, 5, and 6. The third row contains buttons for 7, 8, and 9. The fourth row contains buttons for an empty box, 0, and another empty box. The fifth row contains buttons for a decimal point, a negative sign, and a fraction template icon. Above the grid is a toolbar with icons for undo, redo, left arrow, right arrow, delete, and check.

Other correct responses: for Part A, any equivalent graph; for Part B, any equivalent values

Note: for Part B, the values can be switched

- 17.** A system of inequalities is given.

Inequality 1: $x \geq 4$

Inequality 2: $y > x$

B.E.S.T. Mathematics Sample Items Answer Key

This question has **three** parts.

Part A

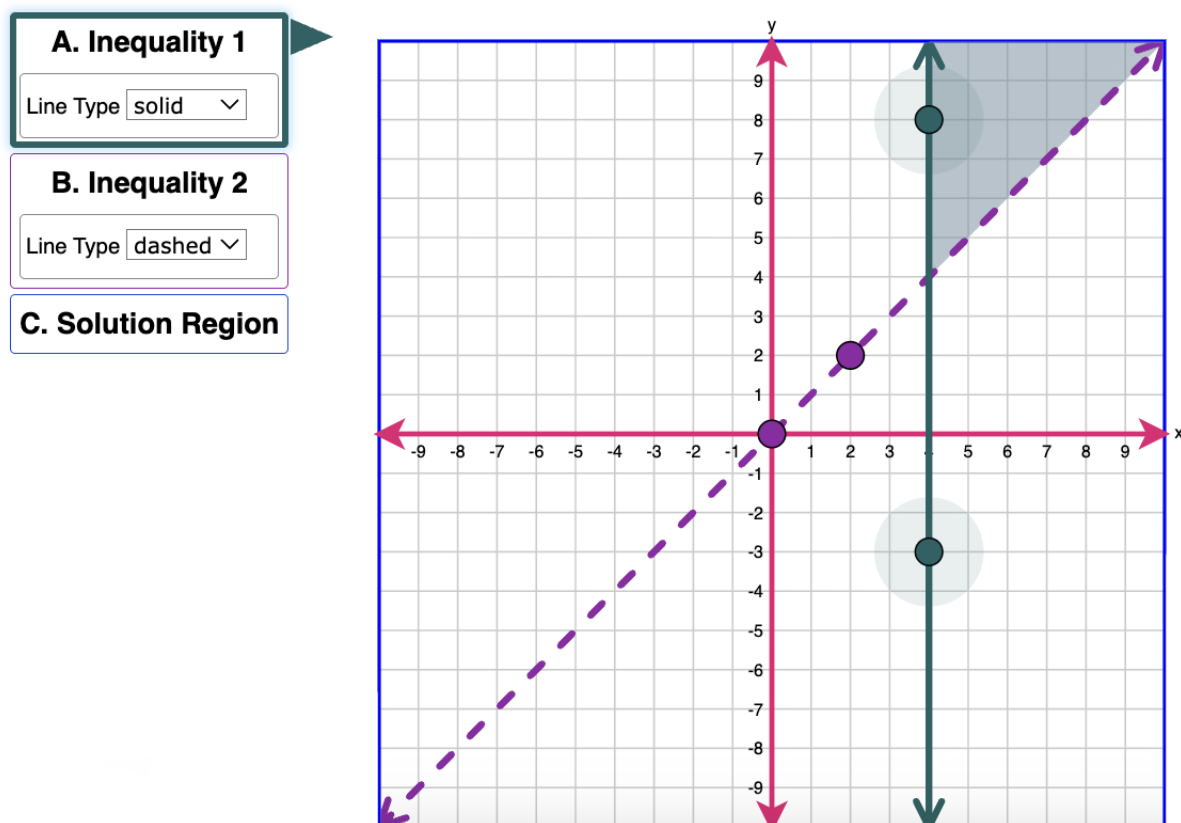
Select a line type and plot points to graph Inequality 1 of the system.

Part B

Select a line type and plot points to graph Inequality 2 of the system.

Part C

Select the region that represents the solution set of the system.



Other correct responses: any graph with the same boundary lines and the correct region shaded

- 18.** A system of inequalities is given.

$$y < x + 4$$

$$y < x + 1$$

The boundary lines of both inequalities are plotted on the coordinate plane.

Select the region or regions on the coordinate plane that represent the solution set of the system of inequalities.

